

**Association of Monocyte/High-Density Lipoprotein ×
Albumin ratio with Acute Kidney Injury following
Off Pump Coronary Artery Bypass Grafting in
Diabetic Patients**



Investigator

Dr. Naima Tullah Hazera

Master of Surgery (Cardiovascular & Thoracic surgery)

Phase B

Reg. No: 18102020291418

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Department of Cardiac Surgery

National Institute of Cardiovascular Diseases, Dhaka

In Affiliation with

Bangabandhu Sheikh Mujib Medical University, Dhaka

GUIDE

Dr. Md. Shafiqul Islam

MBBS

MS (Cardiovascular & Thoracic surgery)

Associate Professor,

Department of Cardiac Surgery,

National Institute of Cardiovascular Diseases (NICVD),

Sher-E-Bangla, Nagar, Dhaka, Bangladesh

Co-guide

Dr. Sakila Israt Jahan

MBBS

MS (Cardiovascular & Thoracic surgery)

Assistant Professor, Department of Paediatric Cardiac Surgery,

National Institute of Cardiovascular Diseases (NICVD),

Sher-E-Bangla Nagar, Dhaka, Bangladesh.

DECLARATION

This primary data-based thesis will be submitted in partial fulfillment of the requirement of the final part examination of Master of Surgery (Cardiovascular & Thoracic Surgery). The research work is going to be done during 1st July, 2023 to 30th June, 2025 at the Department of Cardiac surgery in National Institute of Cardiovascular Diseases (NICVD), Sher-E-Bangla Nagar, Dhaka, Bangladesh. No portion of the work referred to in this thesis protocol has been submitted in support of an application for another degree or any other institution of learning.

Date

Dr. Naima Tullah Hazera

FORWARDING

Certified that Dr. Naima Tullah Hazera is going to carry out the research titled "Association of Monocyte/High-Density Lipoprotein $\times$ Albumin ratio with Acute Kidney Injury following Off Pump Coronary Artery Bypass Grafting in Diabetic Patients" and prepared this thesis under my direct supervision. I have found the work and the thesis protocol satisfactory for partial fulfillment of the requirements of the thesis part examinations of Master of Surgery (Cardiovascular & Thoracic Surgery).

Dr. Md. Shafiqul Islam

MBBS

MS (Cardiovascular & Thoracic surgery)

Associate Professor,

Department of Cardiac Surgery,

National Institute of Cardiovascular Diseases (NICVD),

Sher-E-Bangla, Nagar, Dhaka, Bangladesh

Date:

To

The Director and Professor,

National Institute of Cardiovascular Diseases (NICVD),

Sher E Bangla Nagar, Dhaka, Bangladesh

Subject: Application for approval of thesis protocol

Sir,

With due respect, I do hereby submit my thesis protocol for your kind approval. Prof. Dr. Md. Shafiqul Islam, Associate Professor, Department of Cardiac Surgery, has given the consent to guide me. The title of my thesis is "Association of Monocyte/High-Density Lipoprotein $\times$ Albumin ratio with Acute Kidney Injury following Off Pump Coronary Artery Bypass Grafting in Diabetic Patients”.

I therefore, hope that you would be kind enough to approve my thesis protocol and allow me to work in this institution to complete thesis in due time and oblige thereby.

Sincerely yours,

Dr. Naima Tullah Hazera

Master of Surgery (Cardiovascular & Thoracic surgery), Phase B

Session: March 2023

National Institute of Cardiovascular Diseases, Dhaka.

LIST OF ABBREVIATIONS

ABG= Arterial Blood Gas

ACT= Activated Clotting Time

AF= Atrial Fibrillation

AKI= Acute Kidney Injury

AKIN= Acute Kidney Injury Network

BMI= Body Mass Index

CABG= Coronary Artery Bypass Grafting Surgery

CKD= Chronic Kidney Disease

CPB = Cardiopulmonary bypass

DM = Diabetes Mellitus

eGFR= Estimated Glomerular Filtration Rate

FFP= Fresh Frozen Plasma

Hb= Hemoglobin

HDL= High Density Lipoprotein

HF= Heart Failure

Hct = Hematocrit

HTN = Hypertension

KDIGO= Kidney Disease Improving Global Outcomes Criteria

MHAR= Monocyte/(High-Density Lipoprotein × Albumin) Ratio

NICVD = National Institute of Cardiovascular Diseases

NYHA = New York Heart Association

OPCAB= Off-Pump Coronary Artery Bypass

PC= Platelet Count

ROC= Receiver Operating Characteristic

RRT= Renal Replacement Therapy

SPSS= Statistical Package for Social Sciences

UFH= Unfractionated Heparin

WB= Whole Blood

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INTRODUCTION

Acute kidney injury (AKI) is a sudden loss of kidney function defined by an acute increase in serum creatinine concentration and decrease in urinary output (Kellum and Lameire, 2013). Depending on the various definitions of AKI, 8% to 15% of patients developed AKI after off pump coronary artery bypass grafting (OPCAB) surgery (Ryden *et al.*, 2012).

The incidence of AKI following OPCAB surgery in diabetic patient ranges from 10% to 20% (Shen, *et al.*, 2022). The interplay between malnutrition and inflammation in diabetic patients has been increasingly recognized as a critical factor in the pathogenesis of AKI, warranting comprehensive investigation to elucidate predictive markers and improve clinical outcomes (Singh, *et al.* 2022).

Diabetes mellitus is a global health concern, characterized by chronic hyperglycemia that leads to a spectrum of vascular and metabolic complications. Diabetic patients undergoing OPCAB represent a high-risk cohort due to their pre-existing conditions, including endothelial dysfunction, microvascular disease, and susceptibility to infections. Malnutrition in this population exacerbates these risks, impacting wound healing, immune response, and overall recovery. However, malnutrition in diabetic patients often goes unnoticed because the focus is primarily on strict blood glucose management, overshadowing nutritional deficiencies (Khanimov I, *et al.*, 2019). Symptoms like weight loss, muscle wasting, or micronutrient deficiencies (e.g., vitamin D, B12) may be mistaken for diabetes-related complications, leading delayed diagnosis and management, thus highlighting the need for proactive nutritional assessment and intervention (Karatas *et al.*, 2018).

Inflammation is a well-documented contributor to the development of AKI. During surgical procedures like OPCAB, inflammatory responses are activated as part of the body's defense mechanism. However, excessive inflammation can result in endothelial damage, increased vascular permeability, and subsequent renal hypo perfusion, leading to AKI. In diabetic patients, the baseline inflammatory state is already heightened due to persistent hyperglycemia, which further predisposes them

to exaggerated inflammatory responses during and after surgery, increasing the risk of renal impairment (Singh *et al.*, 2022). Over 30 biomarkers have been reported in recent works (Jefferies *et al.*, 2016). Among them, the usefulness of neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule-1 (KIM-1), cystatin-C (CysC), liver type fatty acid-binding protein (L-FABP), interleukin-6 (IL-6), interleukin-18 (IL-18) to predict AKI within 4-12 hours after cardiac surgery has been shown in many studies (Sugimoto *et al.*, 2017). A study in our settings showed the efficacy of serum CysC as a biomarker to predict AKI 12 hours after CPB. Another recent study in our setting revealed that elevated preoperative NLR was significantly associated with early postoperative AKI in patients undergoing isolated OPCAB (Nahar *et al.*, 2024; Chowdhury *et al.*, 2018). However, most of these markers are too expensive to be used routinely in our country and often fail to account for the interplay between malnutrition and inflammation, particularly in diabetic patients. Therefore, search for new biomarkers for early prediction of AKI in diabetic patients is a highly targeted area of research.

Recently the monocyte-to-HDL ratio (MHR) has emerged as a novel and potent marker of systemic inflammation and metabolic health. Monocytes, as pivotal components of the innate immune system, play a vital role in initiating and sustaining inflammatory responses. HDL, traditionally known for its cardio protective properties, also exerts anti-inflammatory and antioxidant effects, modulating the inflammatory process. An elevated MHR indicates a state of heightened inflammation and diminished anti-inflammatory capacity, reflecting a precarious balance that may predispose individuals to postoperative complications, including AKI (Huang *et al.*, 2021).

Albumin, the most abundant protein in plasma (3.5 g/dL to 5 g/dL), is a negative acute-phase reactant, a critical marker of nutritional status and systemic inflammation. Inflammatory states, such as those encountered in surgical patients, lead to decreased synthesis and increased catabolism of albumin, resulting in hypoalbuminemia. Moreover, hypoalbuminemia is associated with proteinuria in patients with diabetic nephropathy (stage 1 & 2). A recent single center prospective comparative cross-sectional study conducted in Bangladesh revealed, preoperative hypoalbuminemia (Serum albumin < 4.0 g/dL) was associated with AKI in 36.25% patients following OPCAB, bringing poor outcomes in those patients (Salahuddin *et al.*, 2024).

The albumin-to-globulin ratio ((AGR) has traditionally been used to assess the balance between nutritional status and inflammation (Zhang *et al*, 2020). However, combining albumin with MHR, an already used marker provides a more comprehensive measure for both nutritional and inflammatory dimensions, offering a more accurate predictor of adverse outcomes like AKI (Li *et al.*, 2023).

The monocyte/(HDL × albumin) [(M/(H × A))] ratio, a novel and composite marker (MHAR), integrates the dual impact of malnutrition and inflammation on renal function in diabetic patients undergoing OPCAB. The MHAR has recently demonstrated to effectively predict AKI and other adverse effects following cardiac surgery, though the cutoff value varied among different studies. Using a cut-off value of 4.08 , a single center retrospective cohort study revealed that preoperative elevated MHAR was associated with postoperative AKI in 15.7% diabetic patients undergoing OPCAB, with a sensitivity of 89.63% and specificity of 88.34% (Singh *et al.*, 2022).

To the best of our knowledge, to this date, no studies have been carried out in Bangladesh to see the predictive value of MHAR for AKI following OPCAB. The aim of this study is to investigate whether elevated preoperative MHAR is associated with AKI following OPCAB in diabetic population so that we would be able to identify the patients at risk and take prophylactic measures to mitigate potential dangers. We will also attempt to find out a cutoff value of preoperative MHAR to predict post OPCAB AKI in diabetic patients with accuracy in our population.

Chapter 2

RATIONALE

Coronary artery bypass grafting (CABG) surgery, even when performed using OPCAB techniques, is associated with a significant risk of post operative AKI in diabetic patients, a complication that can lead to adverse outcomes, including increased morbidity, prolonged hospitalization, and elevated healthcare costs. Among the potential risk factors, malnutrition and systemic inflammation have emerged as key contributors to postoperative AKI particularly in diabetic patients. Early prediction of postoperative AKI using practical and reliable biomarkers is important for taking necessary precautions to prevent or retard its progression. Despite advancements in perioperative care, current predictive tools for AKI often fail to account for the interplay between malnutrition and inflammation and are too expensive to be used routinely in our country.

While simple and cost effective markers of inflammation, such as monocyte counts, nutritional and anti-inflammatory markers such as, serum albumin and high-density lipoprotein (HDL), have been independently studied, the combined predictive value of these factors have not been extensively explored. The Monocyte/High-Density Lipoprotein $\times$ Albumin $[(M/(H \times A))]$ ratio (MHAR) offers a promising alternative that integrates inflammatory and nutritional status and may provide a more robust tool for identifying patients at risk of AKI following OPCAB in diabetic patients.

To the best of our knowledge, to this date, no studies have been carried out in Bangladesh to see the predictive value of MHAR for AKI following OPCAB. This study will attempt to establish the association between preoperative MHAR and AKI following OPCAB in diabetic patients, so that effective and timely measures can be taken to mitigate potential dangers concerning AKI improving overall surgical outcomes.

Chapter 3

HYPOTHESIS

3.1 RESEARCH HYPOTHESIS

High preoperative monocyte/high-density lipoprotein $\times$ albumin ratio is associated with acute kidney injury following off-pump coronary artery bypass grafting surgery in diabetic patients.

3.2 STATISTICAL HYPOTHESIS

3.2.1 Null hypothesis (H_0)

High preoperative monocyte/high-density lipoprotein $\times$ albumin ratio is not associated with acute kidney injury following off-pump coronary artery bypass grafting surgery in diabetic patients.

3.2.2 Alternate hypothesis (H_A)

High preoperative monocyte/high-density lipoprotein $\times$ albumin ratio is associated with acute kidney injury following off-pump coronary artery bypass grafting surgery in diabetic patients.

OBJECTIVES

4.1 General objective:

To determine association between preoperative monocyte/(high-density lipoprotein × albumin) ratio and acute kidney injury following OPCAB surgery in diabetic patients.

4.2 Specific objective:

- 1) To measure preoperative monocyte/(high-density lipoprotein × albumin) ratio in diabetic patients undergoing off-pump coronary artery bypass grafting and divide them in two groups.
- 2) To detect postoperative AKI in diabetic patients undergoing OPCAB surgery (according to serum creatinine criteria of kidney disease improving global outline 2012 guideline) with normal preoperative monocyte/(high-density lipoprotein × albumin) ratio ($\text{MHAR} < 4.08$),
- 3) To detect postoperative AKI in diabetic patients undergoing OPCAB surgery (according to serum creatinine criteria of kidney disease improving global outline 2012 guideline) with high preoperative monocyte/(high-density lipoprotein × albumin) ratio ($\text{MHAR} \geq 4.08$).
- 4) To compare the incidence of postoperative AKI between two groups.

METHODOLOGY

5.1 Study design:

Prospective observational Study

5.2 Period of study:

1st July, 2023 to 30th June, 2025

5.3 Place of study:

The study will be conducted in the Department of Cardiac Surgery, NICVD, Sher-e-Bangla Nagar, Dhaka, Bangladesh.

5.4 Study population:

Admitted diabetic patients in the Dept. of Cardiac Surgery, NICVD, who will undergo OPCAB surgery.

5.5 Sampling method:

Purposive sampling.

5.6 Ethical Consideration:

- Approval for the study will be obtained from the Ethical Review Committee of NICVD, Sher-e-Bangla Nagar, Dhaka.
- Informed written consent will be taken from each patient or legal guardian.
- Confidentiality of all participants will be maintained strictly.
- The study population will have the full right to withdraw himself/ herself from the study at any time during the study period.
- Data for the study will be collected in an approved data collection form/ sheet.

5.7 Inclusion criteria:

- Adult diabetic patients (18 to 69 years) requiring elective OPCAB.

5.8 Exclusion criteria

- Patients with preoperative renal dysfunction (eGFR < 30 mL/min/1.73 m²)
- Use of IABP in preoperative period
- Patients exhibiting systemic conditions potentially associated with hypoalbuminemia—hepatic dysfunction, active malignancies, various endocrine disorders (like hypo- and hyperthyroidism)
- Those undergoing prolonged steroid therapy or chemotherapy
- Emergency CABG

5.9 Sample size:

Calculated from following equation:

$$n = \frac{P_1(1-P_1) + P_2(1-P_2)}{(P_1 - P_2)^2} \times (Z_\alpha + Z_\beta)^2$$

Where, n = sample size,

Z_α = The Z-score for the desired significance level (e.g., for a 5% significance level, 1.96)

Z_β = the Z-score for the desired power (for 80% power, 0.84)

P_1 = proportion of AKI in patients with serum albumin levels ≥ 4.0 gm/dl = 18.75% = 0.1875 (Salahuddin, *et al.*, 2024)

P_2 = proportion of AKI in patients with serum albumin levels < 4.0 gm/dl = 36.25% = 0.3625 (Salahuddin *et al.*, 2024)

$$\begin{aligned} \text{So, } n &= \frac{0.1875(1-0.1875) + 0.3625(1-0.3625)}{(0.1875-0.3625)^2} \times (1.96 + 0.84)^2 \\ &= 97.6591 \approx 98 \end{aligned}$$

98 patients will be included in each group.

5.10 Grouping of patients:

- Group A = 98 patients undergoing OPCAB with preoperative MHAR < 4.08
- Group B = 98 patients undergoing OPCAB with preoperative MHAR $\geq$ 4.08

5.11 Measures of variables

A. Dependent variables:

Primary outcome variables:

- Acute kidney injury (according to serum creatinine criteria of kidney disease improving global outline 2012 guideline)

Secondary outcome variables:

- Need for renal replacement therapy
- Duration of mechanical ventilation (hours)
- Low cardiac output syndrome
- Duration of ICU stay (days)
- Duration of postoperative hospital stay (days)

B. Independent variables

- Preoperative monocyte/high-density lipoprotein $\times$ albumin ratio

C. Confounding variables

- **Comorbidities:**
 - Hypertension
 - Dyslipidemia
- **Demographic variables:**
 - Age
 - Sex (M/F)
 - Body mass index (kg/m²)
- **Preoperative:**
 - Serum creatinine (mg/dl)
- **Per-operative:**
 - Duration of surgery (in min)
 - Urine output (ml/hour)

5.12 Study procedure

Patients admitted in department of cardiac surgery for OPCAB would be enrolled, provided they meet inclusion and exclusion criteria.

After selection, the aim, objectives and procedure of the study will be explained in details to the subject and a written informed consent will be taken from each subject before enrollment.

The particulars of the patient will be recorded. Detailed history taking and physical examination will be performed following the standard procedure of clinical methods.

Upon enrollment, routine hematological (using BECMAN COULTER Hematology Analyzer, Model- DxH560), biochemical (using BECMAN COULTER Chemistry Analyzer, Model- AU480), microbiological and radiological tests will be performed and patients will be divided into two groups on the basis of preoperative monocyte/(high-density lipoprotein $\times$ albumin) ratio (MHAR) level.

Group A: MHAR < 4.08

Group B: MHAR $\geq$ 4.08

Baseline serum creatinine level will be measured before OPCAB surgery. Preoperative well glycemic control will be achieved.

Pharmacological measures, hemodynamic management and pre-anesthetic checkup will be done according to institutional protocol.

5.13 Anesthesia

All patient will receive a standard general anesthesia protocol. All patients will be intubated for surgery and will have arterial lines, central venous lines and Foley's catheter placed.

Clinical monitoring will routinely include ECG, arterial pressure, central venous pressure, pulse oximetry, urine output monitoring.

5.14 Surgical technique

In the planned procedure, a midline sternotomy will be performed to allow for the harvesting of the left internal mammary artery and saphenous venous grafts. Heparin will be administered at a dose of 200 IU/kg of body weight to achieve an activated clotting time of >300 seconds. Once the desired clotting time is reached, the octopus evolution tissue stabilizer (Medtronic, Inc., Minneapolis, Minnesota, United States) will be placed to stabilize the diseased coronary artery, and intracoronary shunts will be inserted. The distal anastomosis will be completed using 7–0 or 8–0 prolene sutures, while the proximal anastomosis will be performed with 6–0 prolene sutures. Following the completion of the coronary anastomosis, protamine will be administered at a ratio of 1 mg/100 IU of heparin to reverse anticoagulation. Patients will then be transferred to the cardiac ICU for postoperative care and monitoring.

To manage glucose homeostasis, blood glucose levels will be monitored at hourly intervals using arterial blood gas measurements. Perioperative glucose levels will be controlled with insulin infusion, aiming for a target range of 140 to 180 mg/dL.

Mean blood pressure will be maintained above 60 mm Hg.

5.15 Postoperative evaluation:

At the end of the surgery, the patients will be transferred to the Intensive Care Unit (ICU) and managed accordingly as institutional protocol. Antibiotics and analgesics will also be used as per institutional protocol.

After surgery blood samples for serum creatinine will be collected at 24 hours, 48 hours, 7th postoperative day. Postoperative data from the first 7 days will be analyzed to determine the incidence of de novo AKI. AKI will be defined based on the KDIGO 2012 guidelines as any of the following criteria:

- An increase in serum creatinine by ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) within 48 hours,
- An increase in serum creatinine to ≥ 1.5 times the baseline level, known or presumed to have occurred within the last 7 days.

5.16 Statistical Analysis:

The statistical package for the social sciences program (Version 27) will be used to evaluate all data. Chi-Square (χ^2) test, Fisher's exact test & student's *t*-test will be used to analyze data. Quantitative data will be expressed as mean and standard deviation (SD) and comparison between groups will be done by student's *t*-test. Qualitative data will be expressed as frequency and percentage and comparison will be carried by Chi-Square (χ^2) test or Fisher's exact test. To evaluate the efficacy of preoperative MHAR as a biomarker for early detection of postoperative AKI, a receiver operating characteristic (ROC) curve will be generated, and the area under the curve (AUC) will be calculated along with its 95% confidence interval (CI). An AUC value of 1.0 indicates a perfect biomarker, while a value of 0.5 suggests it is no better than random chance. AUC values ranging from 0.7 to 0.8 are deemed acceptable, 0.8 to 0.9 are considered excellent, and values above 0.9 are regarded as outstanding. For all analytic tests, the level of significance set at 0.05 and a p-value of <0.05 will be considered significant. The summarized data will be presented in the form of tables and charts.

Chapter 6

Results

Suitable tables, graphs, charts etc. will show results of the study.

Chapter 7

Discussion

Discussion will be made after obtaining the results in the study according to objectives with references and cross reference.

Chapter 8

Summary

Summary will be drawn after discussion.

Chapter 9

Conclusion

Conclusion will be made upon the results and discussion with the recommendation for future project.

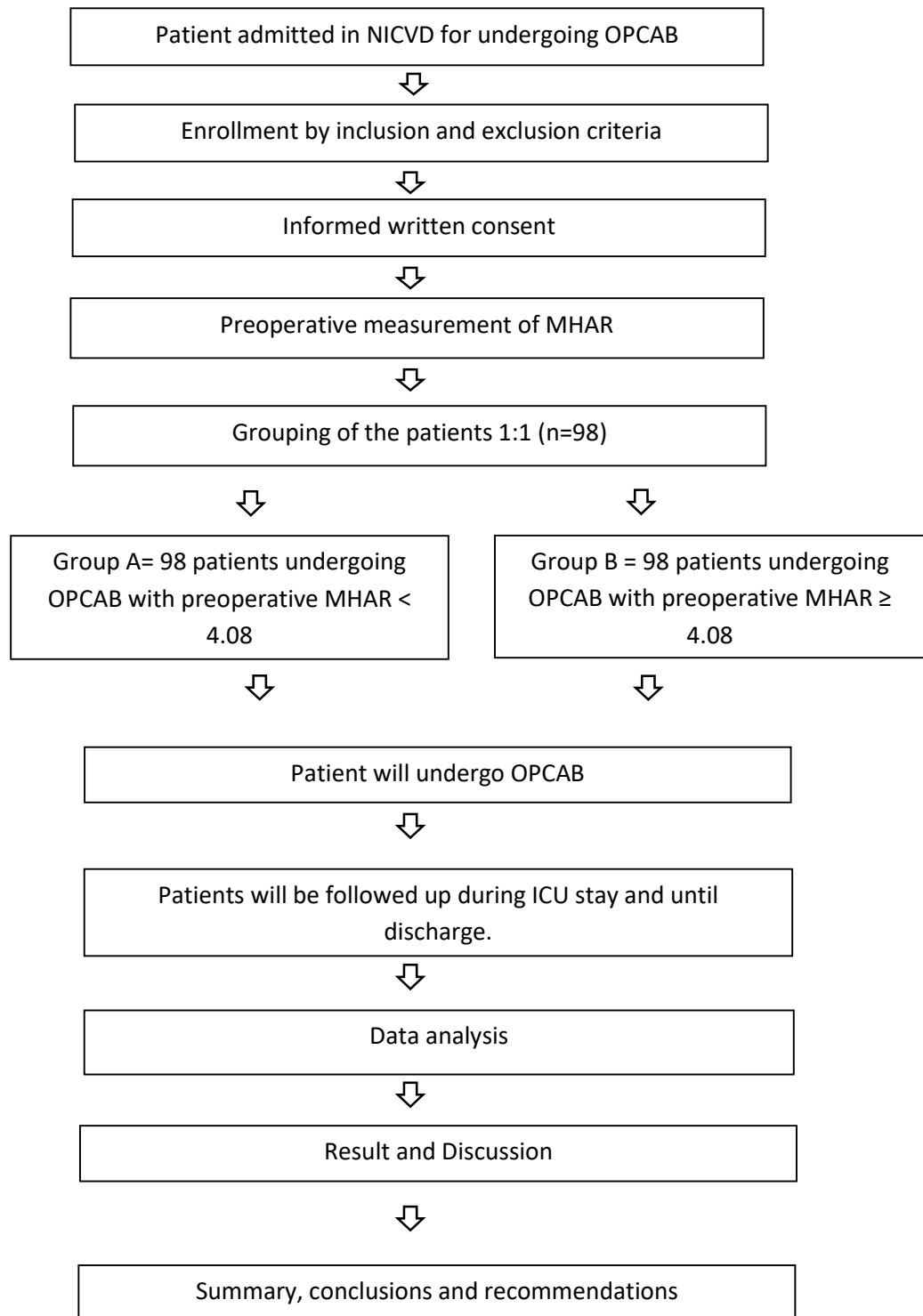
Chapter 10

Recommendation

Based on conclusion.

Chapter 11

STUDY FLOW CHART



Chapter 12

BUDGET PLANNING

A tentative budgetary plan for this study is presented in the table below:

Serial No	Particulars	Amount (Taka)
2	Investigation expenses	200000
4	Stationery	50000
5	Personal & Miscellaneous cost	52000
	Total	302000

Chapter 13

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Appendix I

STUDY DEFINITIONS

Monocyte/[High-Density Lipoprotein (HDL) × Albumin] Ratio (MHAR):

Is a biomarker used to evaluate the balance between inflammation and nutritional status in patients. It combines the levels of monocytes (white blood cell involved in inflammation) and HDL (high density lipoprotein) with serum albumin (a protein that indicates nutritional status). This ratio helps in assessing the risk of adverse outcomes, such as acute kidney injury (AKI), particularly in patients undergoing cardiac surgery (Singh *et al.*, 2022, p. 2).

Acute kidney injury: AKI was defined as the any of the following according to KDIGO 2012 guideline:

- Increase in serum creatinine by ≥ 0.3 mg/dL (≥ 26.5 μ mol/L) within 48 hours;
or
- Increase in serum creatinine to ≥ 1.5 times baseline, which is known or presumed to have occurred within the last 7 days (Fliser et al.,2012)

Diabetes mellitus:

Diabetes mellitus was defined as-

- a. Hemoglobin A1c $\geq 6.5\%$; or
- b. A history of diabetes diagnosed and/or treated by a healthcare provider
- c. Definitions in prior specifications included:
 - i. Fasting plasma glucose ≥ 126 mg/dL (7.0 mmol/L); or
 - ii. Two-hour plasma glucose ≥ 200 mg/dL (11.1 mmol/L) during an oral glucose tolerance test; or
 - iii. In a patient with classic symptoms of hyperglycemia or hyperglycemic crisis, a random plasma glucose ≥ 200 mg/dL (11.1 mmol/L) (Bojar 2020, p. 875).

Estimated glomerular filtration rate (eGFR):

eGFR was calculated using the 2005 Modification of Diet in Renal Disease formula.

$$eGFR = 186.3 \times S \text{ Creatinine (mg/dL)}^{-1.154} \times \text{age (in year)}^{-0.203} \times 0.742 \text{ [if female]} \times 1.212 \text{ [if black]}$$
 (Palevsky et al., 2013, p 108).

Dyslipidemia

According to NCEP ATP-III criteria, dyslipidemia was defined as-

- History of dyslipidemia diagnosed and/or treated by a physician.
- LDL cholesterol > 100 mg/dl.
- Total cholesterol > 200 mg/dl.
- HDL cholesterol < 40 mg/dl.
- Triglycerides > 150 mg/dl. (Stone, *et al.*, 2014).

Hypertension:

Classification of Office Blood Pressure and definitions of Hypertension in adults

Category	Systolic (mmHg)		Diastolic (mmHg)
Normal	<130	And	<85
High Normal	130-139	and/or	85-89
Grade 1 Hypertension	140-159	and/or	90-99
Grade 2 Hypertension	≥160	and/or	≥110

(Unger *et al.*, 2020)

Body mass index (BMI):

Body mass index is a simple calculation using a person's height in meters and weight in kilograms. The formula is $BMI = \text{weight in kg} / \text{height in m}^2$

Normal Weight 18.5 - 22.9kg /m²

Overweight 23 - 24.9kg /m²

Obese > 25kg / m² (Thandassery *et al.*, 2014)

Off-pump coronary artery bypass grafting (OPCAB):

Defined as coronary artery bypass graft surgery on beating heart during which complete revascularization should be achieved with the avoidance of CPB. Deep pericardial sutures and various retraction devices are used to position the heart for grafting without hemodynamic compromise. A stabilizing platform minimizes movement at the site of the arteriotomy. Intracoronary or aortocoronary shunting can minimize ischemia after an arteriotomy is performed (Bojar 2020, p. 14).

Low cardiac output syndrome (LCOS): in this study LCOS is defined as inadequate cardiac output characterized by cardiac index (CI) less than 2.0L/min/m², a PCWP or PAD pressure over 20 mmHg and a heart rate over 100 bpm resulting in inadequate urine output and cold peripheries (Bojar 2020, p.444).

Preoperative renal dysfunction: Renal dysfunction is commonly defined as a serum creatinine (Scr) >1.5 mg/dL or >130 µmol/L (1 mg/dL= 88 µmol/L). A GFR < 60 mL/min/1.73 m² is the best definition of renal dysfunction, and is the level below which there is an increased risk of worsening renal dysfunction and increased mortality. The stages of CKD are defined as follows (in mL/min/1.73 m²).

Stage 1	>90
Stage 2	60–89
Stage 3A	45–59
Stage 3B	30–44
Stage 4	15–29
Stage 5	<15 or on dialysis (Bojar, 2020, p.682).

On-pump coronary artery bypass grafting (ONCAB):

Defined as coronary artery bypass graft surgery performed through a median sternotomy incision with use of CPB where myocardial preservation is provided by cardioplegic arrest (Bojar 2020, p. 13).

Intensive care unit (ICU) stay:

Defined as the duration of ICU stay needed after operation including the potential need for additional stay after readmission (Bojar, 2010, p. 878).

Postoperative hospital stay:

Defined as a duration of postoperative stay of patients in ICU and ward. Normally it is ≤ 7 days, when the discharge criteria are made. However, if there is indication for discharge on or before 7th day, yet the patient prefers to prolong his/her stay for personal convenience then it will not be considered as a prolonged hospital stay (Bojar, 2010, p. 877).

Group	A / B
Patient no.	

Appendix II

QUESTIONNAIRE

Department of cardiac surgery, NICVD, Dhaka.

Study Name: Association of Monocyte/High-Density Lipoprotein $\times$ Albumin Ratio with Acute Kidney Injury following Off Pump Coronary Artery Bypass Grafting in Diabetic Patients.

Section 1 : General information of the patient

Serial No:		Hospital Reg. No:
Name:		Sex:
Age:	Occupation:	Contact No:
Address:		Unit:
Ward No:	Bed No:	Date of Admission:
Date of Operation:		Date of Discharge:

Section 2: preoperative findings

Preoperative demographic and clinical findings:

Age (years)		
Sex	Male	Female
BMI (kg/m ²)		
HTN	Yes	No
Dyslipidemia	Yes	No

Preoperative laboratory findings:

Serum creatinine (mg/dl)	
Monocyte count (/mm ³)	
Serum HDL (mg/dl)	
Serum albumin (g/dl) Serum albumin (g/dl)	
Monocyte/(high-density lipoprotein × albumin) ratio	

Section 3: per-operative variables:

Duration of surgery (hours)	
Urine output (ml/hour)	

Section 4: postoperative variables

Serum creatinine (mg/dl)	
24 hours after surgery	
48 hours after surgery	
7 days after surgery	
Postoperative acute kidney injury	Yes No
Need for renal replacement therapy	Yes No
Duration of mechanical ventilation (hours)	
Duration of ICU stay (days)	
Duration of postoperative hospital stay (days)	

Appendix III

Consent form

After being fully informed about the objective, consequence of the study “**Association of Monocyte/High-Density Lipoprotein $\times$ Albumin ratio with Acute Kidney Injury following Off Pump Coronary Artery Bypass Grafting in Diabetic Patients**” and my/my patient’s right to withdraw my patient/myself from the study at any time for any purpose, what so ever,

I am,, / on behalf of my patient, hereby giving consent to participate in the study conducted by Dr. Naima Tullah Hazera, MS (Cardiovascular & Thoracic Surgery) Phase B, National Institute of Cardiovascular Diseases (NICVD), Dhaka.

I fully recognize that my/ my patient’s participation in this study will generate valuable medical information that might be used for the interest of patients in future.

Hospital authority, Doctors and any other Stuff will not be responsible for any adverse consequences during the study.

Signature

Thumb Impression

Date.....

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Signature of Witness

Signature of investigator

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Date.....

Date

Appendix IV

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□□□□□ **“Association of Monocyte/High-Density Lipoprotein × Albumin ratio
with Acute Kidney Injury following Off Pump Coronary Artery Bypass Grafting in
Diabetic Patients”** □□□□□□□□ □□□□□□□□□ □□□□□□□□□

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